Lesson 6.4 Guided Notes

Session 1 · Writing Numerical Expressions

I can write a numerical expression that matches a situation. · 6.EE.A.2

Word / Spanish	Meaning — complete the blank	Example
numerical expression / expresión numérica	Numbers and operations with no _____ sign.	4 × 9 + 3
evaluate / evaluar	Find the one _____ an expression equals.	4 × 9 + 3 = 39
product / producto	The answer when you _____.	the product of 6 and 7 is 6 × 7

Key idea

Equal groups (“each,” “per”) in a story usually mean _____.

A total of different groups means _____ the products.

“Less than” flips the order: 5 less than 12 is _____ − 5.

Worked example

A team buys 6 shirts at $7 each and 1 hat for $5. Write an expression for the total cost. Evaluate it.

Shirts are equal groups: 6 × _____.

Add the one hat: 6 × 7 + _____.

Evaluate: 42 + 5 = _____.

Answer: 6 × 7 + 5 = $_____

You try

1. Write a numerical expression.
a. the product of 6 and 4, plus 2 _____
b. 10 less than 35 _____

_____

2. Jo buys 3 books at $5 each and a $4 bookmark. Write an expression for the total cost. Evaluate it.

_____

Sum it up: To write an expression for a story, I look for ___ because ___.

_____

Session 2 · Order of Operations with Powers

I can evaluate an expression that contains grouping symbols and powers using the order of operations. · 6.EE.A.2c

Word / Spanish	Meaning — complete the blank	Example
order of operations / orden de las operaciones	The rules for which _____ to do first, next and last.	( ), powers, × ÷, + −
grouping symbols / símbolos de agrupación	Parentheses or brackets; do the work _____ them first.	(9 − 7)
left to right / de izquierda a derecha	Do partner steps (× and ÷, or + and −) in the order they _____.	24 ÷ 4 × 2 = 6 × 2 = 12

Key idea

First do work inside _____ symbols, then evaluate _____.

Next multiply and divide from _____ to right.

Last, add and subtract from left to _____.

Worked example

Evaluate $40 - (2 + 3)^2 \div 5$.

Parentheses: $2 + 3 =$ _____, so $40 - 5^2 \div 5$.

Power: $5^2 =$ _____, so $40 - 25 \div 5$.

Divide, then subtract: $40 -$ _____.

Answer: $40 - (2 + 3)^2 \div 5 =$ _____

You try

1. Evaluate.
a. $12 - 2 \times 5 =$ _____
b. $(3 + 1)^2 =$ _____

_____

2. Evaluate $6 + 2^3 \times 3$.

_____

Sum it up: In $2 + 3 \times 4^2$, I start with ___ because ___.

_____

Session 1 · Model, reason, and explain

Writing Numerical Expressions · 6.EE.A.2

Success: represent the quantities, justify the relationship, and check the result with the correct units.

A club buys 4 boxes with 6 markers each and 3 single markers. Match the story to 4 × 6 + 3 or 4 × (6 + 3). Explain the difference.

Model the relationship	Name the parts	Explain the structure	Check the result

_____

_____

_____

Extend your reasoning

Create a contrasting example that exposes the misconception: groups a one-time amount with a repeated amount.. Explain what changes and what stays true.

_____

_____

Self-check: Did my model, calculation, explanation, and units agree?

Session 2 · Model, reason, and explain

Order of Operations with Powers · 6.EE.A.2c

Success: represent the quantities, justify the relationship, and check the result with the correct units.

Compare 2×3^2 and $(2 \times 3)^2$. Then evaluate $24 \div 4 \times 2$. Explain the relevant base and equal-precedence rule.

Model the relationship	Name the parts	Explain the structure	Check the result

_____

_____

_____

Extend your reasoning

Create a contrasting example that exposes the misconception: applies the power to the wrong base or performs all multiplication before division.. Explain what changes and what stays true.

_____

_____

Self-check: Did my model, calculation, explanation, and units agree?